

The empirical recurrence for OEIS A251145, proved by transfer matrix

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Abstract

OEIS A251145 counts 5-column arrays over $\{0, \dots, 2\}$ in which every 2×2 subblock sums to one of $\{4\}$, and in which no 2×2 subblock has exactly two nonzero entries, and carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true, and it is not an empirical matter at all: the rows of such an array are the vertices of a finite digraph in which the admissible row-to-row steps are the edges, so $a(n) = \mathbf{1}^\top M^n \mathbf{1}$ for the adjacency matrix M of that digraph, on $S = 243$ vertices. Writing q for the characteristic polynomial of the conjectured recurrence, the recurrence holds for every n exactly when $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$ for all $j \geq 0$, and the Cayley–Hamilton theorem bounds that to $j < S$. It is therefore a finite computation in exact integer arithmetic, and it comes out zero.

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1 The sequence and the conjecture

OEIS A251145 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with every 2×2 subblock summing to 4 and no 2×2 subblock having exactly two nonzero entries.”. It has offset 1 and begins

$$107, 161, 247, 401, 667, 1141, 1987, 3521, \dots$$

The entry carries the following comment:

$$\text{Empirical: } a(n) = 3a(n-1) - 5a(n-3) + a(n-4) + 2a(n-5).$$

As of the “Last modified” line on the live entry (Nov 26 2018 11:23:38, revision 16) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 3a(n-1) - 5a(n-3) + a(n-4) + 2a(n-5) \tag{1}$$

for all $n > 5$, which is every n at which a recurrence of order 5 can be stated.

2 The rows form a digraph

Let $V = \{0, 1, \dots, 2\}^5$, the set of possible rows, so $|V| = S = 243$. For $r, s \in V$ declare $r \rightarrow s$ an edge when, for every j with $1 \leq j \leq 5-1$,

- $r_j + r_{j+1} + s_j + s_{j+1} \in \{4\}$;
- the four entries $r_j, r_{j+1}, s_j, s_{j+1}$ do not include exactly two nonzero ones, that is $\#\{v \in \{r_j, r_{j+1}, s_j, s_{j+1}\} : v \neq 0\} \neq 2$;

that is, when placing the row s directly under the row r satisfies the entry's condition on every 2×2 subblock formed by those two rows. Let $M \in \{0,1\}^{S \times S}$ be the adjacency matrix of this digraph and $\mathbf{1}$ the all-ones vector.

Lemma 1. *The arrays counted by the entry with $n+1$ rows are exactly the walks $r^{(0)} \rightarrow r^{(1)} \rightarrow \dots \rightarrow r^{(n)}$ of length n in this digraph. Consequently*

$$a(n) = \mathbf{1}^\top M^n \mathbf{1}.$$

Proof. An array with $n+1$ rows and 5 columns is precisely a sequence $r^{(0)}, \dots, r^{(n)}$ of elements of V . Every 2×2 subblock of the array lies in two vertically adjacent rows $r^{(i)}, r^{(i+1)}$ and in two horizontally adjacent columns $j, j+1$; so the array satisfies the entry's condition on all of its 2×2 subblocks if and only if each consecutive pair $(r^{(i)}, r^{(i+1)})$ satisfies it on all column pairs, which is exactly the edge relation. Counting walks of length n by summing over all starting and ending vertices gives $\mathbf{1}^\top M^n \mathbf{1}$, since $(M^n)_{r,s}$ is the number of walks of length n from r to s . $\square$

3 The criterion

Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of (1), with the c_i read off that equation.

Lemma 2. *For every $n \geq 5$,*

$$a(n) - \sum_i c_i a(n-i) = \mathbf{1}^\top M^{n-5} q(M) \mathbf{1}.$$

Hence (1) holds for every $n \geq 5$ if and only if $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$ for every $j \geq 0$; and it is enough to check $j = 0, 1, \dots, S-1$.

Proof. By Lemma 1, $a(n-i) = \mathbf{1}^\top M^{n-i} \mathbf{1}$, so

$$a(n) - \sum_i c_i a(n-i) = \mathbf{1}^\top \left(M^n - \sum_i c_i M^{n-i} \right) \mathbf{1} = \mathbf{1}^\top M^{n-5} \left(M^5 - \sum_i c_i M^{5-i} \right) \mathbf{1},$$

which is the stated identity. The equivalence follows by letting $n-5 = j$ run over ≥ 0 . For the last clause put $w = q(M) \mathbf{1}$. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so $\mathbf{1}^\top M^j w$ for $j \geq S$ is the corresponding combination of the earlier values; if those all vanish, so do all the rest. $\square$

4 The computation

Theorem 3. *The recurrence (1) holds for every $n > 5$. This is the statement of Section 1.*

Proof. The vector $w = q(M) \mathbf{1} \in \mathbb{Z}^S$ was formed by 5 matrix–vector products in exact integer arithmetic, and $\mathbf{1}^\top M^j w$ was evaluated for $j = 0, 1, \dots$ in exact integer arithmetic. Every one of those integers is 0 from $j = 5 - 5 + 1$ onwards, and the run was continued past S consecutive zeros, which by Lemma 2 settles every larger j as well. Hence the recurrence holds for every $n > 5$. $\square$

Remark 4. The argument gives more than the recurrence: it shows the sequence is $\mathcal{C}$ -finite with characteristic polynomial dividing that of M , so it has a rational generating function whose denominator divides $\det(I - xM)$. The conjectured recurrence is one consequence; the minimal one is a divisor of q .

5 Verification

Three checks, each independent of the algebra above.

First, the digraph was built directly from the entry’s stated condition and the walk counts $\mathbf{1}^\top M^n \mathbf{1}$ compared against the entry’s DATA at every one of the 29 published indices; the values agree exactly. That is what ties the digraph to the sequence: a model that reproduced only some of the published terms would be the wrong model, and would have been rejected.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was run to the full Cayley–Hamilton bound $j < S = 243$ rather than to a sampled prefix, in exact integer arithmetic, and returned zero throughout.

References

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